РОССИЙСКИЙ УНИВЕРСИТЕТ ДРУЖБЫ НАРОДОВ

Факультет физико-математических и естественных наук

Кафедра прикладной информатики и теории вероятностей

ОТЧЁТ

ПО ЛАБОРАТОРНОЙ РАБОТЕ №5

Продвинутое использование git

Выполнил: Дмитраков Михаил Алексеевич

Москва 2026

Цель работы

Получение навыков правильной работы с репозиториями git.

Теоретические сведения

Gitflow Workflow опубликована и популяризирована Винсентом Дриссеном. Gitflow Workflow предполагает выстраивание строгой модели ветвления с учётом выпуска проекта. Данная модель отлично подходит для организации рабочего процесса на основе релизов. Работа по модели Gitflow включает создание отдельной ветки для исправлений ошибок в рабочей среде.

Последовательность действий при работе по модели Gitflow:

- Из ветки master создаётся ветка develop.
- Из ветки develop создаётся ветка release.
- Из ветки develop создаются ветки feature.
- Когда работа над веткой feature завершена, она сливается с веткой develop.
- Когда работа над веткой release завершена, она сливается в ветки develop и master.
- Если в master обнаружена проблема, из master создаётся ветка hotfix.
- Когда работа над веткой hotfix завершена, она сливается в ветки develop и master.

Общепринятые коммиты

Стандарт Conventional Commits — это простое соглашение о том, как следует формировать сообщение коммита. Он описывает простой набор правил для создания понятной истории коммитов, а также позволяет проще разрабатывать инструменты автоматизации, основанные на истории коммитов.

Выполнение лабораторной работы

Работа с тестовым репозиторием

Установка Node.js

Установил Node.js с помощью пакетного менеджера (рис. 1).

```
dmitrakovma@dmitrakovma:~$ pnpm setup
Appended new lines to /home/dmitrakovma/.bashrc


Next configuration changes were made:
export PNPM_HOME="/home/dmitrakovma/.local/share/pnpm"
case ":$PATH:" in
  *":$PNPM_HOME:"*) ;;
  *) export PATH="$PNPM_HOME:$PATH" ;;
esac


To start using pnpm, run:
source /home/dmitrakovma/.bashrc
dmitrakovma@dmitrakovma:~$ source ~/.bashrc
dmitrakovma@dmitrakovma:~$
```

Настройка commitizen

Выполнил настройку commitizen для помощи в форматировании коммитов (рис. 2).

```
dmitrakovma@dmitrakovma:~$ pnpm add -g commitizen

Update available! 10.27.0 → 10.30.1.
Changelog: https://pnpm.io/v/10.30.1
To update, run: pnpm self-update

WARN 2 deprecated subdependencies found: glob@7.2.3, inflight@1.0.6
Packages: +149
++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
Progress: resolved 149, reused 0, downloaded 149, added 149, done

/home/dmitrakovma/.local/share/pnpm/global/5:
+ commitizen 4.3.1

Done in 2.5s using pnpm v10.27.0
```

Настройка standard-changelog

Выполнил настройку standard-changelog для автоматического создания логов (рис. 3).

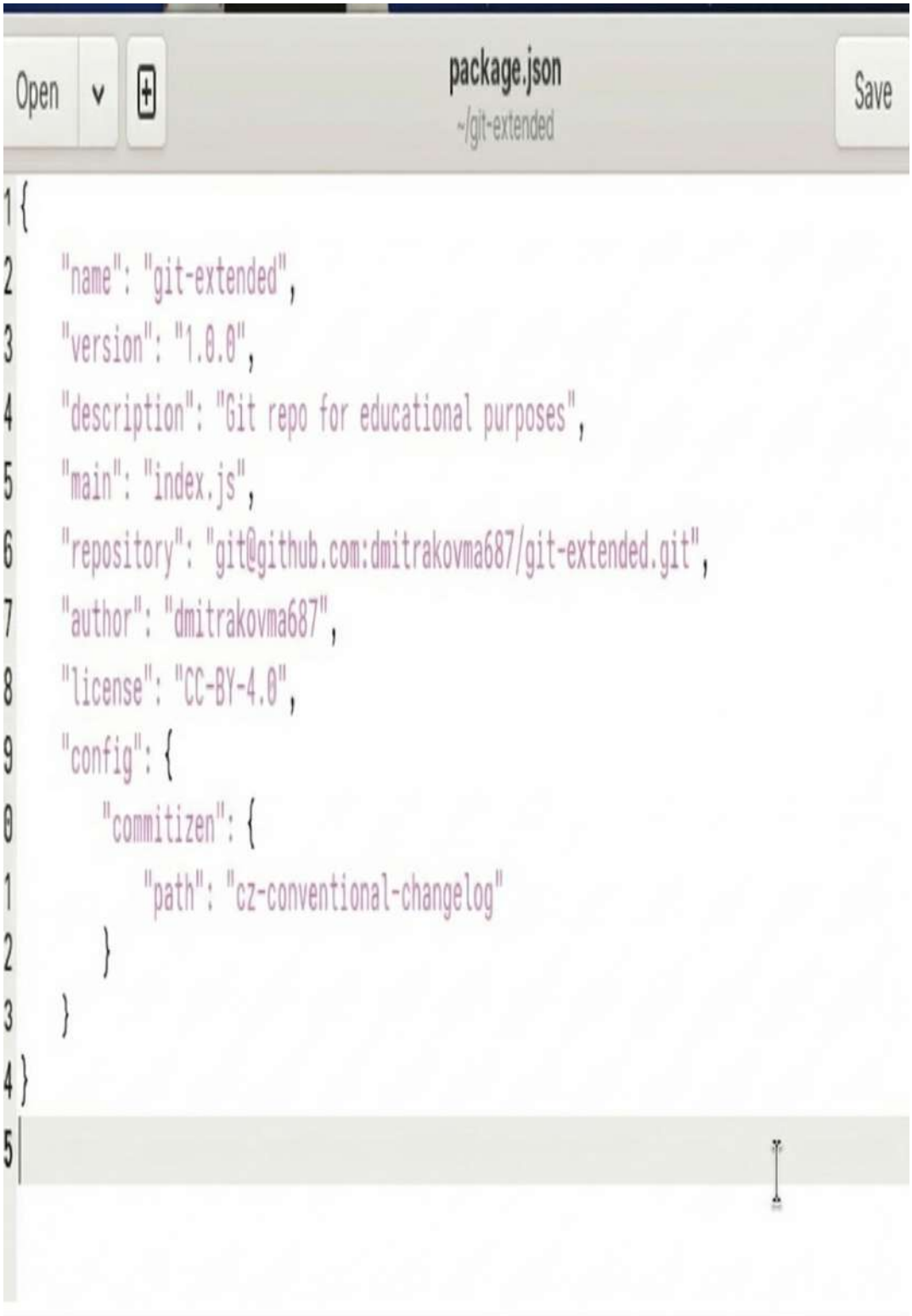
```
dmitrakovma@dmitrakovma:~$ pnpm add -g standard-changelog
WARN 2 deprecated subdependencies found: glob@7.2.3, inflight@1.0.6
Packages: +32
++++++++++++++++++++++++++++++++
Progress: resolved 181, reused 149, downloaded 32, added 32, done

/home/dmitrakovma/.local/share/pnpm/global/5:
+ standard-changelog 7.0.1

Done in 2.4s using pnpm v10.27.0
```

Создание package.json

Создал файл package.json с необходимой конфигурацией для работы с conventional commits (рис. 4).



Использование commitizen

Выполнил коммит с помощью commitizen, используя команду cz-conventional-changelog (рис. 5).

```
dmitrakovma@dmitrakovma:~/git-extended$ git add .
dmitrakovma@dmitrakovma:~/git-extended$ git cz
cz-cli@4.3.1, cz-conventional-changelog@3.3.0

? Select the type of change that you're committing: chore:   Other changes tha
modify src or test files
? What is the scope of this change (e.g. component or file name): (press enter

? Write a short, imperative tense description of the change (max 93 chars):
 (4) test
? Provide a longer description of the change: (press enter to skip)

? Are there any breaking changes? No
? Does this change affect any open issues? No
[master b3869cd] chore: test
 1 file changed, 15 insertions(+)
 create mode 100644 package.json
dmitrakovma@dmitrakovma:~/git-extended$ git push
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 1.12 KiB | 1.12 MiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:dmitrakovma687/git-extended.git
   4837057..b3869cd  master -> master
dmitrakovma@dmitrakovma:~/git-extended$
```

Инициализация git-flow

Инициализировал git-flow в тестовом репозитории (рис. 6).

```
dmitrakovma@dmitrakovmaa:~/git-extended$ git flow init

Which branch should be used for bringing forth production releases?
   - master
Branch name for production releases: [master]
Branch name for "next release" development: [develop]

How to name your supporting branch prefixes?
Feature branches? [feature/]
Bugfix branches? [bugfix/]
Release branches? [release/]
Hotfix branches? [hotfix/]
Support branches? [support/]
Version tag prefix? [] v
Hooks and filters directory? [/home/dmitrakovma/git-extended/.git/hooks]
dmitrakovma@dmitrakovma:~/git-extended$ git branch
* develop
  master
dmitrakovma@dmitrakovma:~/git-extended$ git push --all
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'develop' on GitHub by visiting:
remote:      https://github.com/dmitrakovma/git-extended/pull/new/develop
remote:
To github.com:dmitrakovma/git-extended.git
 * [new branch]      develop -> develop
dmitrakovma@dmitrakovma:~/git-extended$
```

Завершение релиза

Завершил релиз в тестовом репозитории с помощью git-flow (рис. 7).

```
dmitrakovma@dmitrakovma:~/git-extended$ git push --all
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 8 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (5/5), 2.68 KiB | 2.68 MiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:dmitrakovma687/git-extended.git
   b3869cd..b0bbe5d  develop -> develop
   b3869cd..2de5666  master -> master
dmitrakovma@dmitrakovma:~/git-extended$ git push --tags
Enumerating objects: 1, done.
Counting objects: 100% (1/1), done.
Writing objects: 100% (1/1), 159 bytes | 159.00 KiB/s, done.
Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:dmitrakovmna687/git-extended.git
 * [new tag]         v1.0.0 -> v1.0.0
dmitrakovma@dmitrakovma:~/git-extended$ gh release create v1.0.0 -F CHANGELOG
https://github.com/dmitrakovmma687/git-extended/releases/tag/v1.0.0
dmitrakovma@dmitrakovma:~/git-extended$
```

Отправка на GitHub

Отправил изменения на GitHub, включая все ветки и теги (рис. 8).

```
dmitrakovvma@dmitrakovvma:~/git-extended$ git flow feature start feature_branch
Switched to a new branch 'feature/feature_branch'

Summary of actions:
- A new branch 'feature/feature_branch' was created, based on 'develop'
- You are now on branch 'feature/feature_branch'

Now, start committing on your feature. When done, use:

     git flow feature finish feature_branch

dmitrakovma@dmitrakovma:~/git-extended$ touch test2
dmitrakovma@dmitrakovma:~/git-extended$ git flow feature finish feature_branc
Switched to branch 'develop'
Your branch is up to date with 'origin/develop'.
Already up to date.
Deleted branch feature/feature_branch (was b0bbe5d).

Summary of actions:
- The feature branch 'feature/feature_branch' was merged into 'develop'
- Feature branch 'feature/feature_branch' has been locally deleted
- You are now on branch 'develop'

dmitrakovma@dmitrakovma:~/git-extended$
```

Объединение веток

Выполнил объединение веток develop и master (рис. 9).

```
mitrakovma@dmitrakovma:~/git-extended$ git flow release start 1.0.0
witched to a new branch 'release/1.0.0'

ummary of actions:
 A new branch 'release/1.0.0' was created, based on 'develop'
 You are now on branch 'release/1.0.0'

ollow-up actions:
 Bump the version number now!
 Start committing last-minute fixes in preparing your release
 When done, run:

    git flow release finish '1.0.0'

mitrakovma@dmitrakovma:~/git-extended$ standard-changelog --first-release
mitrakovma@dmitrakovma:~/git-extended$ git add CHANGELOG.md
mitrakovma@dmitrakovma:~/git-extended$ git commit -am 'chore(site): add ch
release/1.0.0 7541729] chore(site): add changelog
1 file changed, 4 insertions(+)
create mode 100644 CHANGELOG.md
mitrakovma@dmitrakovma:~/git-extended$ git flow release finish 1.0.0
witched to branch 'master'
our branch is up to date with 'origin/master'.
erge made by the 'ort' strategy.
CHANGELOG.md | 4 ++++
1 file changed, 4 insertions(+)
create mode 100644 CHANGELOG.md
lready on 'master'
our branch is ahead of 'origin/master' by 2 commits.
 (use "git push" to publish your local commits)
witched to branch 'develop'
```

Завершение релиза (повтор)

Завершил релиз и отправил результаты на GitHub (рис. 10).

```
Summary of actions:
- Release branch 'release/1.2.3' has been merged into 'master'
- The release was tagged 'v1.2.3'
- Release tag 'v1.2.3' has been back-merged into 'develop'
- Release branch 'release/1.2.3' has been locally deleted
- You are now on branch 'develop'

dmitrakovma@dmitrakovma:~/git-extended$ git push --all
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 8 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (6/6), 2.79 KiB | 2.79 MiB/s, done.
Total 6 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:dmitrakovma687/git-extended.git
   b0bbe5d..a148120  develop -> develop
   2de5666..1b4b2ee  master -> master
dmitrakovma@dmitrakovma:~/git-extended$ git push --tags
Enumerating objects: 1, done.
Counting objects: 100% (1/1), done.
Writing objects: 100% (1/1), 157 bytes | 157.00 KiB/s, done.
Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:dmitrakovma687/git-extended.git
 * [new tag]         v1.2.3 -> v1.2.3
dmitrakovma@dmitrakovma:~/git-extended$ gh release create v1.2.3 -F CHANGELOG
https://github.com/dmitrakovma6687/git-extended/releases/tag/v1.2.3
dmitrakovma@dmitrakovma:~/git-extended$
```

Подготовка рабочего репозитория

Создание package.json и коммит

Создал файл package.json в рабочем репозитории и выполнил коммит (рис. 11).

```
dmitrakovma@dmitrakovma16:~/work/study/2025-2026/Операционные системы/2026-1--st
-intro$ git add .
dmitrakovma@dmitrakovma16:~/work/study/2025-2026/Операционные системы/2026-1--st
-intro$ git cz
cz-cli@4.3.1, cz-conventional-changelog@3.3.0

? Select the type of change that you're committing: chore:   Other changes that
modify src or test files
? What is the scope of this change (e.g. component or file name): (press enter t

? Write a short, imperative tense description of the change (max 93 chars):
 (4) test
? Provide a longer description of the change: (press enter to skip)

? Are there any breaking changes? No
? Does this change affect any open issues? No
[master 4fd6ecd] chore: test
 1 file changed, 11 insertions(+), 11 deletions(-)
dmitrakovma@dmitrakovma16:~/work/study/2025-2026/Операционные системы/2026-1--st
-intro$ git push
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 1.12 KiB | 1.12 MiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:dmitrakovma687/2026-1--study--os-intro.git
   b8b1b3f..4fd6ecd  master -> master
dmitrakovma@dmitrakovma16:~/work/study/2025-2026/Операционные системы/2026-1--st
-intro$
```

Завершение релиза

Завершил релиз в рабочем репозитории (рис. 12).

```
Summary of actions:
- Release branch 'release/1.0.0' has been merged into 'master'
- The release was tagged 'v1.0.0'
- Release tag 'v1.0.0' has been back-merged into 'develop'
- Release branch 'release/1.0.0' has been locally deleted
- You are now on branch 'develop'

dmitrakovma@ddmitrakovmma:~/work/study/2025-2026/Операционные системы/2026-1--s
-intro$ git push --all
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 8 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 2.76 KiB | 2.76 MiB/s, done.
Total 5 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:dmitrakovma687/2026-1--study--os-intro.git
   4fd6ecd..25e7f86  develop -> develop
   4fd6ecd..40fbd8a  master -> master
dmitrakovma@ddmitrakoovma:~/work/study/2025-2026/Операционные системы/2026-1--s
-intro$ git push --tags
Enumerating objects: 1, done.
Counting objects: 100% (1/1), done.
Writing objects: 100% (1/1), 156 bytes | 156.00 KiB/s, done.
Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:dmitrakovma687/2026-1--study--os-intro.git
 * [new tag]         v1.0.0 -> v1.0.0
dmitrakovma@dmitrakovmama:~/work/study/2025-2026/Операционные системы/2026-1--s
-intro$
```

Вывод

В ходе выполнения данной лабораторной работы я получил навыки правильной работы с репозиториями git, освоил модель Gitflow, научился использовать conventional commits с помощью commitizen и standard-changelog.